

DATA 266 Homework 5

DATA 266 — Homework 5

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SID4 = 0670 | SEED = 670 | SLICE = 670 | HP_ID = 4 | CLS_A = 0 | CLS_B = 5

Notebook: lora_dialogsum.ipynb. Dataset: neil-code/dialogsum-test. Model: google/flan-t5-small.

Same SID/SEED as HW1. No extra HP_ID network. The two trained models are LoRA r=4 and LoRA r=16.

I froze the base weights and trained a small update on the q and v layers ($\Delta W = B \times A$). Alpha 16, dropout 0.05, both ranks.

1. Data

Input is the dialogue with the prefix `summarize the dialogue:.` Target is the summary column.

I trained on 256 rows from the train split, shuffled with seed 670. The two dialogues I compare are validation ids dev_56 and dev_357, so they are not in that 256.

Two processed samples are in the notebook (HR department, and the apartment call that ends at 19 Lingual Road).

2. Baseline

No adapters. 76,961,152 parameters, all of them the original model.

id	What it printed
dev_56	Li Na's mother has been building a fire under her since her neighbour's daughter got married.
dev_357	Adam, I'm sorry. I'm sorry.

First one copies a line. Second one just repeats the apology.

3. LoRA

	r=4	r=16
alpha	16	16
dropout	0.05	0.05
total params	77,133,184	77,649,280

	r=4	r=16
trainable	172,032 (0.223%)	688,128 (0.886%)

4. Training

2 epochs, batch 4, lr 1e-3, on MPS. About a minute each.

Rank	Mean train loss	Last logged loss (step 128)
r=4	1.796	1.511
r=16	1.793	1.498

Logged loss starts near 2.46 and ends near 1.50. The two curves sit on top of each other.



Training loss

Adapters: adapters/r4 and adapters/r16.

5–6. Same two dialogues after

dev_56 (reference: Li Na is pressed by her mother about marriage)

- Before: the copied “building a fire” sentence.
- r=4: “Li Na’s mother has been building a fire under her neighbour’s daughter.”
- r=16: “Li Na’s mother has been building a fire under Li Na’s neighbour’s daughter.”

dev_357 (reference: Alice is over an hour late and apologizes)

- Before: “Adam, I’m sorry. I’m sorry.”
- $r=4$: Alice left work late, missed the bus, and her boss asked for urgent letters.
- $r=16$: longer, and it also says someone is over an hour late, but it says Adam is late.

Before fine-tuning the model either copied a line or echoed “I’m sorry.” After LoRA, both ranks try to pack a few facts into one summary. That is why dialogue 2 is actually about being late, not just the apology. Rank 16 writes more on that dialogue, but it swaps who was late.

7. $r=4$ vs $r=16$

$r=16$ has about $4\times$ the trainable weights (688,128 vs 172,032). Alpha is 16 for both, so α/r is 4 when $r=4$ and 1 when $r=16$.

Mean loss is 1.796 vs 1.793. Mean word overlap on the two dialogues is 0.254 vs 0.259. On dialogue 2, $r=16$ overlaps the reference more (0.375 vs 0.286) but adds the Adam/Alice swap. On dialogue 1, $r=16$ is worse (0.143 vs 0.222) because it still copies the idiom and sticks it on the neighbour’s daughter.

With 256 examples and 2 epochs, the bigger rank did not give me a clearly better summary. The small one was enough.